MA104 · DISCRETE MATHEMATICS

Assignment 03 — Mathematical Induction

Questions: **11**
Due: ___ / ___ / **2026**

Chapter 3 — Lecture 3 with the exercise sheets E02 and E03

Name: ..... **ID:** .....

Group: ..... **Date:** .....

Instructions. Every proof must show the **basis step**, the **inductive hypothesis** and the **inductive step**, and mark the line where you use the hypothesis — write "(by the IH)". **Answer without AI**, and do not copy from a classmate or from the solved sheets; you may be asked to explain any answer at the board.

Question 1 cf. L03 Ex. 1

Prove that for all $n \geq 1$,

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2$$

Question 2 cf. L03 Ex. 5

Prove that for all $n \geq 1$,

$$2 + 4 + 6 + \cdots + 2n = n(n + 1)$$

Question 3 E02 Q2 · cf. L03 Ex. 7

Prove that $6^n - 1$ is divisible by 5 for every $n \geq 1$.

Question 4 E02 Q1 · E03 Ex. 5

Prove that for all $n \geq 1$,

$$1 + 4 + 7 + \cdots + (3n - 2) = \frac{n(3n-1)}{2}$$

Question 5 E03 Ex. 2 · cf. L03 Ex. 5

Prove that for all $n \geq 1$,

$$3 + 3^2 + 3^3 + \cdots + 3^n = \frac{3^{n+1}-3}{2}$$

Question 6 E03 Ex. 3

Prove that for all $n \geq 1$,

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}$$

Question 7 E03 Ex. 4

The triangular numbers are $1, 3, 6, 10, \ldots$ with n-th term $\frac{n(n+1)}{2}$. Prove that for all $n \geq 1$,

$$1 + 3 + 6 + 10 + \cdots + \frac{n(n+1)}{2} = \frac{n(n+1)(n+2)}{6}$$

Question 8 cf. L03 Ex. 3, 4

Prove that for all $n \geq 1$,

$$\frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \frac{1}{7 \cdot 10} + \cdots + \frac{1}{(3n-2)(3n+1)} = \frac{n}{3n+1}$$

Question 9 cf. L03 Ex. 7

Prove that $4^n - 1$ is divisible by 3 for every $n \geq 1$.

Question 10 cf. L03 Ex. 7

Prove that $3^{2n} - 1$ is divisible by 8 for every $n \geq 1$.

Question 11 cf. L03 Ex. 6

Prove that $n^3 + 5n$ is divisible by 6 for every $n \geq 1$.